# EROLLA RISHVIN REDDY

Blockchain Developer • Web3 Engineer • Smart Contracts

Solidity • Ethereum • Smart Contracts • IPFS • Web3 • SHA-256 • TypeScript • Node.js

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GitHub Portfolio LinkedIn

## Professional Summary

Computer Science Engineering student specializing in **Blockchain, IoT & Cybersecurity (BIC)** with a focused interest in **Blockchain Development, Web3, Smart Contracts, Decentralized Systems, Cryptographic Verification, and Blockchain Security**. Experienced in engineering blockchain-backed applications using **Ethereum, Solidity, IPFS, SHA-256, TypeScript, Node.js, MQTT, REST APIs, and PostgreSQL**, with practical experience integrating distributed ledger technology with **IoT security and digital forensics**. Founder of **Rishvin Labs**, developing systems for **immutable evidence registration, tamper detection, cryptographic integrity verification, decentralized evidence storage, and blockchain-backed chain-of-custody workflows**. Strong foundation in **smart contract architecture, distributed systems, cryptography, secure software development, database systems, computer networks, and IoT architectures**, with a focus on building practical and verifiable Web3 solutions.

## Education

**Woxsen University** Hyderabad, India
*B.Tech in Computer Science and Engineering* 2024 – 2028
*Specialization: Blockchain, IoT & Cybersecurity (BIC)* CGPA: **9.01/10**

## Blockchain & Web3 Technical Skills

**Blockchain & Web3:** Blockchain Technology, Distributed Ledger Technology (DLT), Ethereum, Web3, Decentralized Applications (dApps), Decentralized Systems, Immutable Records, Blockchain Architecture
**Smart Contracts:** Solidity, Smart Contract Development, Contract Architecture, On-Chain Evidence Registration, Transaction Verification, Contract Events, Blockchain-Based Data Integrity
**Cryptography & Integrity:** SHA-256, Cryptographic Hashing, Data Integrity, Hash Verification, Tamper Detection, Evidence Verification, Cryptographic Provenance, Secure Data Validation
**Decentralized Storage:** IPFS, Content Addressing, Distributed Storage, Hash-Based Retrieval, Decentralized Evidence Storage, Off-Chain Data Preservation
**Blockchain Security:** Smart Contract Security Concepts, Access Control, Evidence Integrity, Immutable Audit Trails, Blockchain-Based Chain of Custody, Secure Transaction Workflows, Blockchain Forensics Concepts
**Blockchain & IoT:** IoT Security, MQTT, ESP32, Device Telemetry, Event-Driven Architecture, IoT Evidence Collection, Blockchain-Backed IoT Systems
**Blockchain Backend Engineering:** TypeScript, JavaScript, Node.js, REST APIs, PostgreSQL, JSON, Docker, API Integration, Event-Driven Services
**Development Tools:** Git, GitHub, Docker, Postman, Linux, Antigravity IDE, VS Code, Overleaf
**Computer Science Foundations:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Software Engineering, System Design, Secure Application Development

## Technical Experience

**Founder & Blockchain / Security Engineer – Rishvin Labs** *2025 – Present*

- Founded and lead **Rishvin Labs**, developing blockchain-backed systems across **Web3, smart contracts, cryptographic verification, IoT security, digital forensics, and decentralized evidence preservation**.
- Architected and developed blockchain-enabled applications using **Ethereum, Solidity, IPFS, SHA-256, TypeScript, Node.js, MQTT, PostgreSQL, REST APIs, and Docker**.
- Engineered decentralized evidence workflows supporting **on-chain evidence registration, cryptographic hash verification, tamper detection, immutable audit trails, and chain-of-custody verification**.
- Designed systems integrating **blockchain networks with IoT telemetry and digital-forensics pipelines**, connecting off-chain evidence with verifiable on-chain integrity records.
- Applied software engineering principles including **smart contract architecture, API integration, event-driven systems, database design, secure communication, testing, debugging, and deployment** to blockchain applications.

**Pega Platform Intern – Pegasystems** *Aug 2026 – Present*

– Selected for the **National Internship Program 2026 with SmartBridge**, gaining hands-on exposure to enterprise application development and workflow automation using the **Pega Platform**.
– Developing practical knowledge of **Business Process Management (BPM), Case Management, Digital Process Automation (DPA), workflow orchestration, application configuration, and enterprise data integration**.
– Applying structured enterprise software engineering practices involving **application lifecycle management, business process modeling, case lifecycle design, data integration, and secure enterprise application development**.

**Blockchain & Decentralized Systems Engineering** *Academic & Independent Engineering*

– Designed and implemented blockchain-backed systems combining **smart contracts, cryptographic hashing, decentralized storage, IoT telemetry, and digital-forensics workflows**.
– Developed practical experience with **Ethereum smart contracts, Solidity-based evidence registration, IPFS content addressing, SHA-256 integrity verification, and transaction-based auditability**.
– Applied blockchain technology to real-world security problems involving **tamper detection, immutable records, evidence provenance, decentralized verification, and chain-of-custody management**.

## Featured Blockchain Projects

**ChainForensics – Blockchain-Based IoT Forensic Evidence Platform** TypeScript, Node.js, MQTT, Ethereum, Solidity, IPFS, SHA-256

- Engineered a **blockchain-backed digital forensics platform** for collecting, verifying, preserving, and investigating security evidence generated by distributed IoT environments.
- Implemented an evidence pipeline using **MQTT event collection, deterministic JSON serialization, SHA-256 hashing, Ethereum smart contracts, and IPFS** to establish verifiable evidence integrity and provenance.
- Developed an **EvidenceRegistry.sol smart contract** for immutable on-chain registration of forensic evidence metadata and cryptographic hashes, enabling independent verification of evidence integrity.
- Designed mechanisms for **tamper detection, immutable audit trails, chain-of-custody verification, transaction-based evidence validation, and forensic timeline reconstruction**.

**Blockchain Evidence Registry – Verifiable Digital Evidence Architecture** Solidity, Ethereum, TypeScript, Node.js, SHA-256, IPFS

- Designed a **verifiable blockchain evidence architecture** connecting off-chain forensic artifacts with cryptographic hashes and immutable on-chain records.
- Implemented workflows for **evidence registration, hash verification, transaction tracking, metadata integrity, and tamper detection** using Ethereum smart contracts.
- Applied a hybrid **on-chain/off-chain storage model**, using blockchain for verifiable integrity records and IPFS for distributed evidence storage.

**IoT Security Evidence Pipeline – Blockchain-Backed Device Telemetry** ESP32, MQTT, Node.js, TypeScript, Ethereum, Solidity, SHA-256

- Designed an **IoT-to-blockchain evidence pipeline** for converting security events from connected devices into cryptographically verifiable forensic records.
- Integrated **MQTT telemetry, event-driven Node.js services, deterministic serialization, SHA-256 hashing, and Ethereum transactions** for trusted evidence registration.
- Applied blockchain technology to **IoT security, event provenance, tamper detection, and distributed auditability** across connected-device environments.

**Secure Biometric Voting System – Verifiable Identity Architecture** TypeScript, React, Node.js, ESP32, Fingerprint Biometrics

- Developing a **secure biometric voting platform** focused on reliable identity verification, controlled access, duplicate-vote prevention, and auditable election workflows.
- Integrated hardware-based fingerprint authentication using a **Mantra fingerprint device** and designed secure communication between biometric hardware and application services.
- Applied principles of **identity verification, access control, data integrity, auditability, and tamper-resistant system design** to a security-sensitive application domain.

**ChainForensics Research Architecture – Decentralized Forensic Chain of Custody** Blockchain, Ethereum, Solidity, IPFS, Cryptography, Digital Forensics

- Explored a **decentralized chain-of-custody architecture** for maintaining verifiable provenance of digital evidence throughout collection, preservation, investigation, and validation stages.
- Applied **cryptographic hashing, immutable ledger records, decentralized storage, and transaction verification** to strengthen forensic evidence integrity.
- Designed the architecture around **IoT event acquisition, evidence hashing, blockchain registration, distributed storage, and forensic investigation**, combining Web3 with cybersecurity and digital forensics.

## Certifications

**Introduction to Cyber Attacks** – New York University (Coursera)
**Career Essentials in GitHub Professional Certificate** – GitHub & LinkedIn Learning
**Practical GitHub Actions** – LinkedIn Learning
**Practical GitHub Code Search** – LinkedIn Learning
**Practical GitHub Project Management and Collaboration** – LinkedIn Learning
**Programming for Everybody (Getting Started with Python)** – University of Michigan (Coursera)
**Data Structures Using Python: An Introduction** – Coursera
**Object-Oriented Design** – University of Alberta (Coursera)
**Java Programming** – Coursera

## Relevant Coursework

**Blockchain & Distributed Systems:** Blockchain Technology, Distributed Systems, Smart Contract Concepts, Cryptographic Systems, Decentralized Architectures
**Security & Cryptography:** Cybersecurity, Information Security, Network Security, Digital Forensics, Secure Communication, Cryptographic Hashing, IoT Security
**Computer Science Foundations:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Object-Oriented Programming, Software Engineering
**Systems & Engineering:** Internet of Things, Computer Networks, System Design, Database Design, Probability & Statistics, Problem Solving

## Patent & Innovation

**Government of India Registered Design Patent – IoT Connectivity Device** *Co-Inventor — Design No: 470097-001*

- Co-inventor of a **Government of India registered design patent** for an IoT connectivity device, demonstrating innovation in **connected-device engineering, IoT infrastructure, and device communication**.
- Contributed to the development of an engineering concept involving **IoT connectivity and connected-system integration**, translating an engineering solution into a formally registered intellectual property asset.